

Redes Convolucionais II

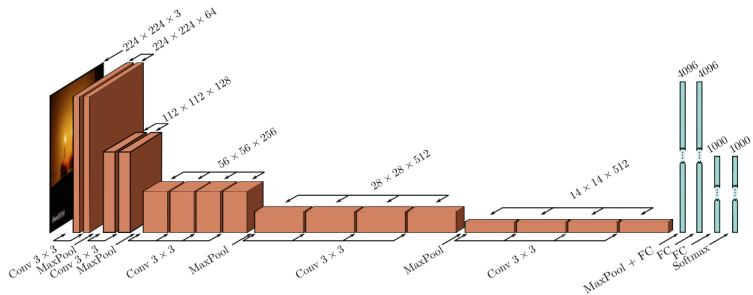
Machine Learning 2 2026.1

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IMPA

April 4, 2026

CNN: Recap



[8] 2016 — Joseph Redmon et al. — *You Only Look Once: Unified, Real-Time Object Detection*

Data Augmentation

- ▶ É uma técnica de pré-processamento para aumentar artificialmente a diversidade do conjunto de dados
- ▶ Cria novas amostras a partir de amostras existentes
- ▶ Melhora a capacidade de generalização e reduz o overfitting

Data Augmentation: Técnicas Comnus

- ▶ Geometric transformations (rotation, translation, scaling)



Flipping

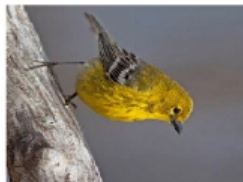


Data Augmentation: Técnicas Comnus

- Geometric transformations (rotation, translation, scaling)



Rotation

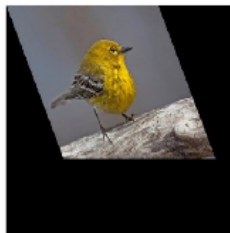


Data Augmentation: Técnicas Comuns

- ▶ Geometric transformations (rotation, translation, scaling)

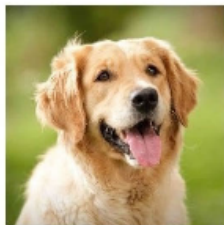


Shearing

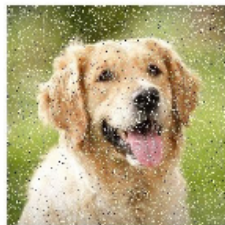


Data Augmentation: Técnicas Comuns

► Noise injection



Noise Injection



Data Augmentation: Técnicas Comuns

► Random erasing



- [5] 2024 — Teerath Kumar et al. — *Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions*
[13] 2017 — Zhun Zhong et al. — *Random Erasing Data Augmentation*

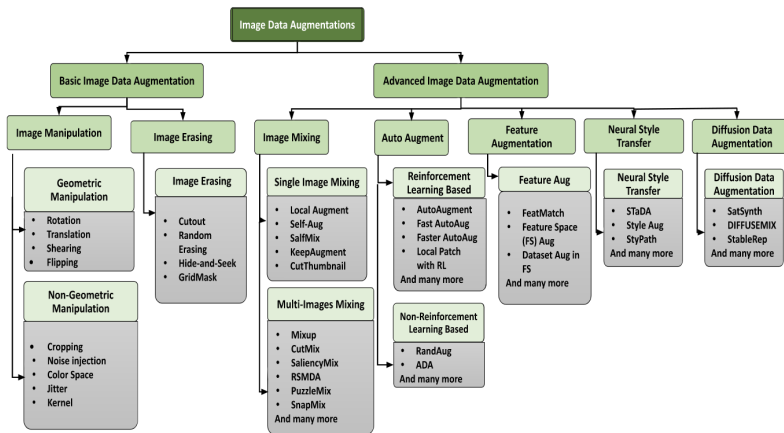
Data Augmentation: Técnicas Comnus

- Mixup, Cutout, CutMix



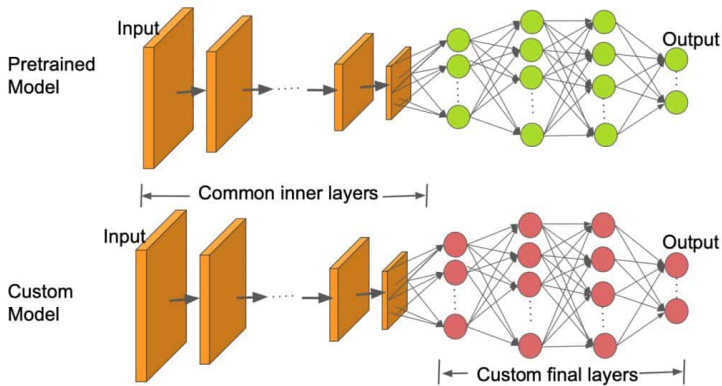
- [5] 2024 — Teerath Kumar et al. — *Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions*
- [13] 2017 — Zhun Zhong et al. — *Random Erasing Data Augmentation*
- [1] 2024 — Pengguang Chen et al. — *GridMask Data Augmentation*
- [12] 2018 — Hongyi Zhang et al. — *mixup: Beyond Empirical Risk Minimization*
- [10] 2019 — Sangdoo Yun et al. — *CutMix: Regularization Strategy to Train Strong Classifiers with Localizable Features*

Data Augmentation: Um panorama



Transfer Learning / Fine-Tuning: Utilizando Modelos Pré-Treinados

- Definição: Usar uma CNN treinada em um grande *dataset* (ex., ImageNet) e adaptá-la para uma nova tarefa.



Transfer Learning / Fine-Tuning: Utilizando Modelos Pré-Treinados

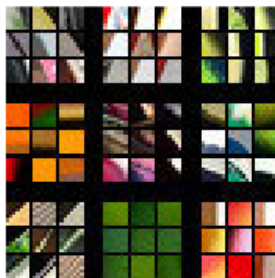
- ▶ Definição: Usar uma CNN treinada em um grande *dataset* (ex., ImageNet) e adaptá-la para uma nova tarefa.
- ▶ Vantagens:
 - ▶ Reduz o tempo de treinamento e os recursos computacionais necessários
 - ▶ Pode melhorar a precisão em tarefas com poucos dados rotulados
- ▶ Estratégias:
 - ▶ Manter as primeiras camadas congeladas e treinar apenas as últimas.
 - ▶ Descongelar todas as camadas e treinar com uma taxa de aprendizado menor.
- ▶ Considerações:
 - ▶ Similaridade de domínio
 - ▶ Tamanho do conjunto de dados
 - ▶ Risco de overfitting

Visualização de Características de uma CNN

- Visualizar a primeira camada é fácil

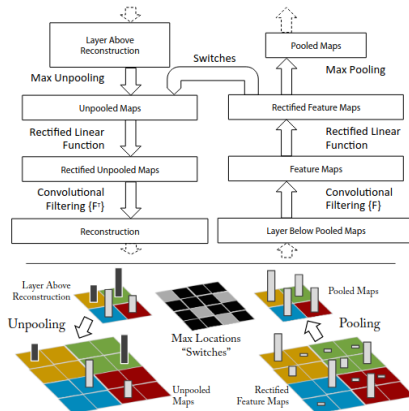


Layer 1



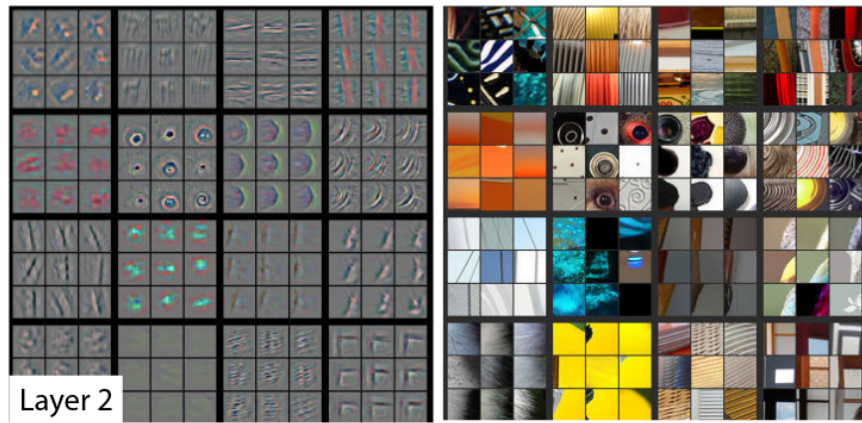
Visualização de Características de uma CNN

► Deconvolutional Networks



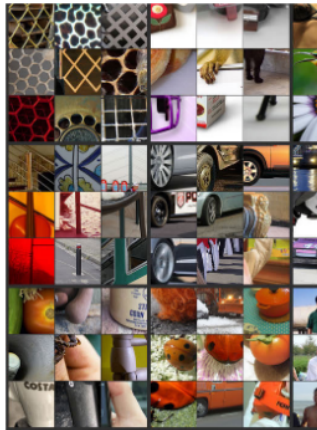
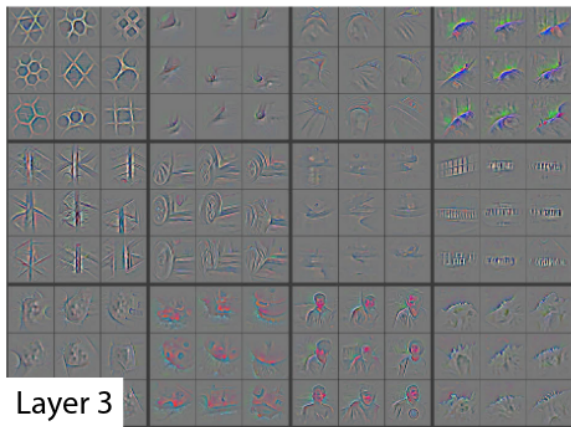
Visualização de Características de uma CNN

► Visualização de Ativação



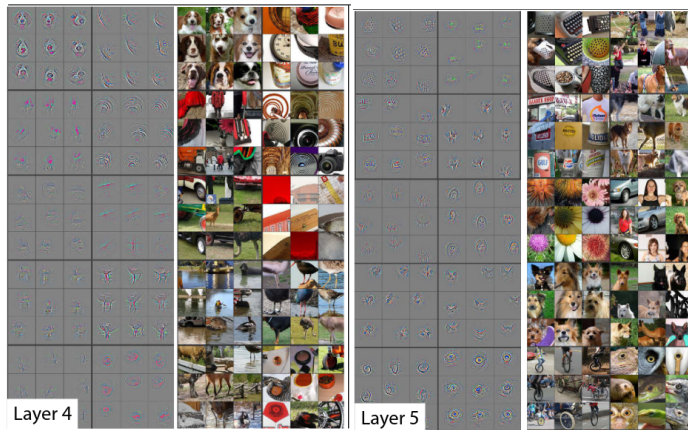
Visualização de Características de uma CNN

► Visualização de Ativação



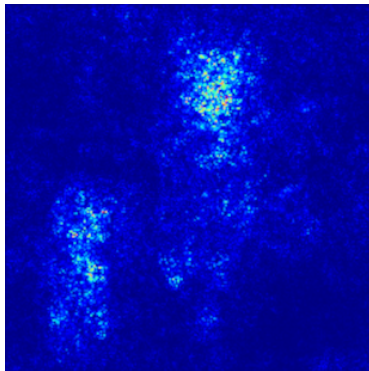
Visualização de Características de uma CNN

► Visualização de Ativação



Saliency Map

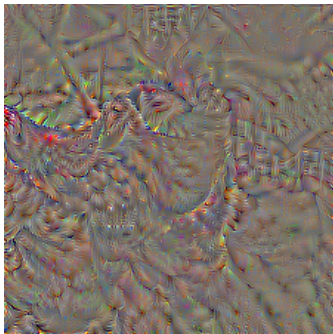
- ▶ Calculando o gradiente da imagem de entrada em relação à ativação de uma camada específica verificamos quais pixels são mais relevantes para a ativação



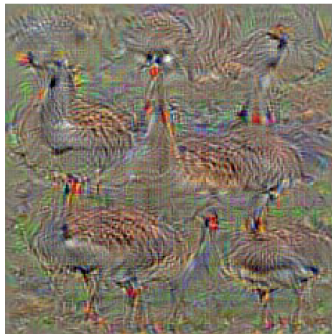
[9] 2014 — Karen Simonyan et al. — *Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps*

Feature Visualization

- ▶ Maximizando o valor de ativação de uma camada podemos gerar imagens
- ▶ $\operatorname{argmax}_I S_c(I) - \lambda \cdot R(I)$, onde $S_c(I)$ é a ativação da camada c e $R(I)$ é uma regularização



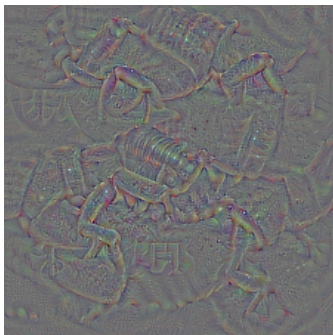
Hen *



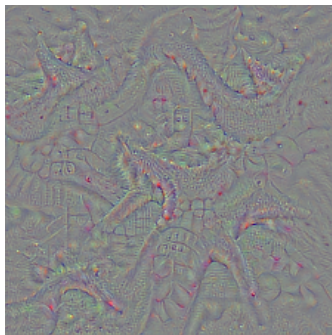
Goose

Feature Visualization

- ▶ Maximizando o valor de ativação de uma camada podemos gerar imagens
- ▶ $\operatorname{argmax}_I S_c(I) - \lambda \cdot R(I)$, onde $S_c(I)$ é a ativação da camada c e $R(I)$ é uma regularização



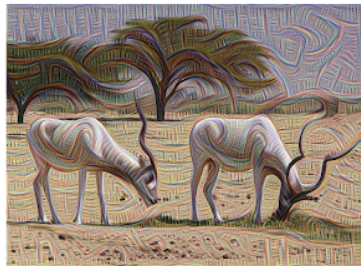
Scorpion *



Star Fish *

Feature Visualization

- ▶ Podemos partir de uma imagem existente e maximizar a ativação de uma camada qualquer

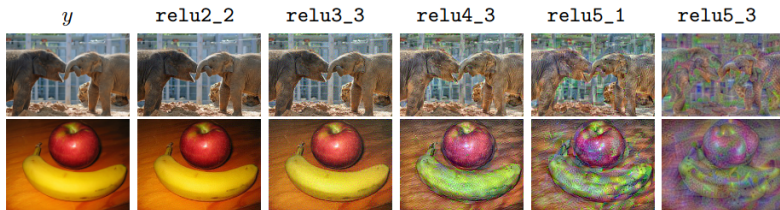


Content Loss



$$\mathcal{L}(p, q, l) = \sum_{i,j} (P_{ij}^l - Q_{ij}^l)^2$$

- onde j é a camada, h, w são posições espaciais e c, c' são canais



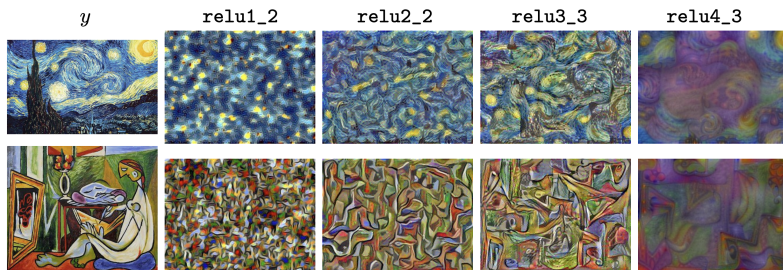
Style Loss



$$G_j^\phi(x)_{c,c'} = \frac{1}{C_j H_j W_j} \sum_{h=1}^{H_j} \sum_{w=1}^{W_j} \phi_j(x)_{h,w,c} \phi_j(x)_{h,w,c'}$$

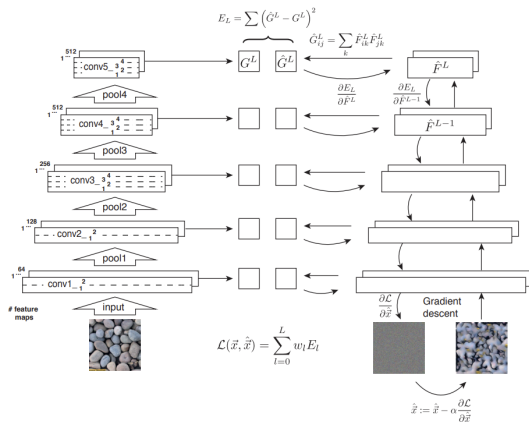
$$\ell_{\text{style}}^{\phi,j}(\hat{y}, y) = \left\| G_j^\phi(\hat{y}) - G_j^\phi(y) \right\|_F^2$$

► onde j é a camada, h, w são posições espaciais e c, c' são canais

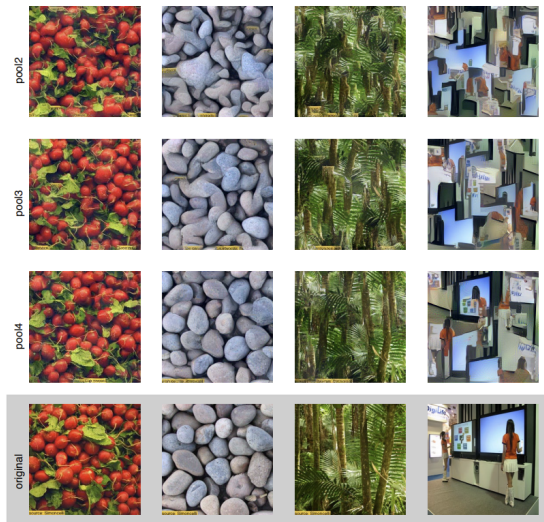


Síntese de Texturas

- ▶ Maximizar as ativações muda a **estrutura** da imagem
- ▶ Podemos, no entanto, maximizar a correlação entre as ativações de diferentes imagens

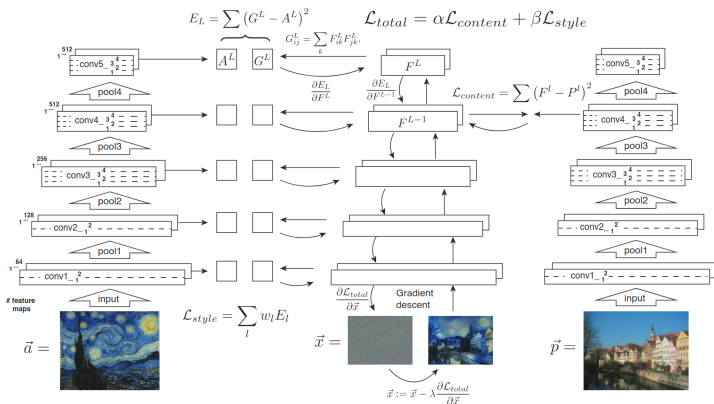


Sintese de Texturas

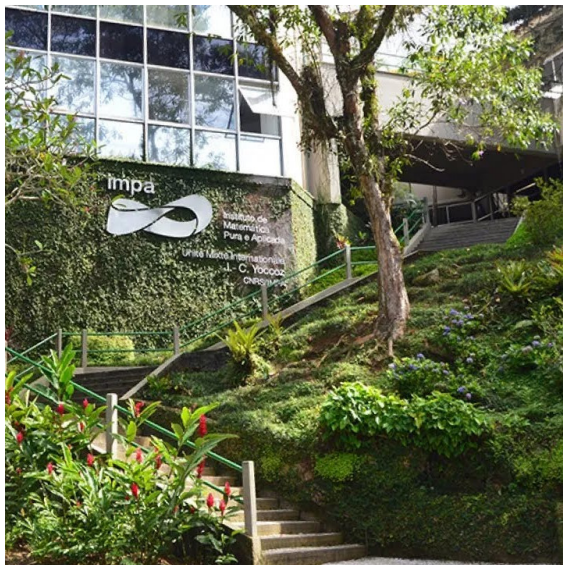


[3] 2015 — Leon A. Gatys et al. — *Texture Synthesis Using Convolutional Neural Networks*

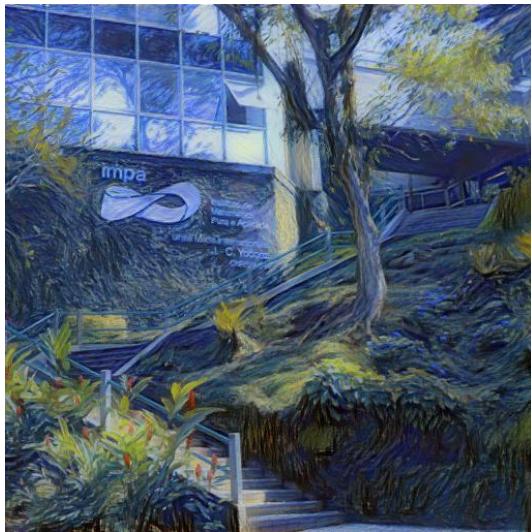
Neural Style Transfer



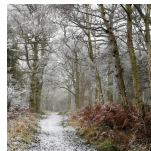
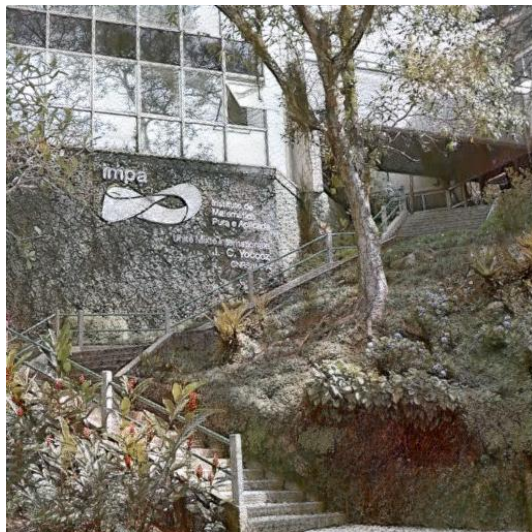
Neural Style Transfer



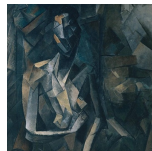
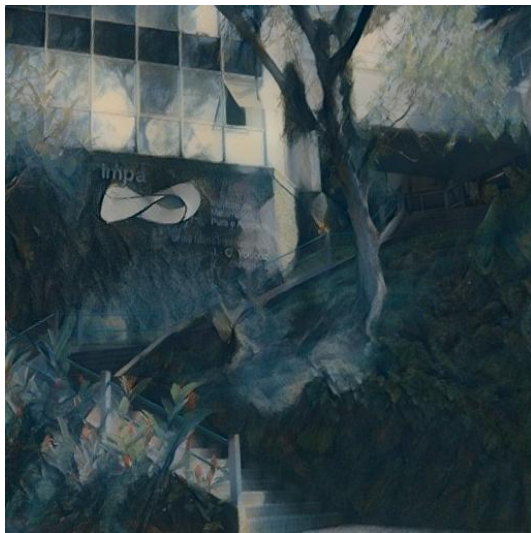
Neural Style Transfer



Neural Style Transfer



Neural Style Transfer



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- [2] Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge. "Image Style Transfer Using Convolutional Neural Networks". In: *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. Vol. 11. 21. IEEE, June 8, 2016, pp. 2414–2423. ISBN: 978-1-4673-8851-1. DOI: 10.1109/CVPR.2016.265. URL: http://openaccess.thecvf.com/content_cvpr_2016/html/Gatys_Image_Style_Transfer_CVPR_2016_paper.html.
- [3] Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge. *Texture Synthesis Using Convolutional Neural Networks*. Read_Status: New Read_Status_Date: 2025-04-04T23:05:15.043Z. Nov. 6, 2015. DOI: 10.48550/arXiv.1505.07376. arXiv: 1505.07376[cs]. URL: <http://arxiv.org/abs/1505.07376> (visited on 04/04/2025).

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- [5] Teerath Kumar et al. "Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions". In: *IEEE Access* 12 (2024). Conference Name: IEEE Access Read_Status: New Read_Status_Date: 2025-04-04T13:56:03.045Z, pp. 187536–187571. ISSN: 2169-3536. DOI: 10.1109/ACCESS.2024.3470122. URL: <https://ieeexplore.ieee.org/document/10699340/?arnumber=10699340> (visited on 04/04/2025).

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- [9] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. *Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps*. Read_Status: New Read_Status_Date: 2025-04-05T02:24:34.965Z. Apr. 19, 2014. DOI: 10.48550/arXiv.1312.6034. arXiv: 1312.6034[cs]. URL: <http://arxiv.org/abs/1312.6034> (visited on 04/05/2025).
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- [12] Hongyi Zhang et al. *mixup: Beyond Empirical Risk Minimization*. Read_Status: New Read_Status_Date: 2024-09-13T19:05:46.429Z. Apr. 27, 2018. DOI: 10.48550/arXiv.1710.09412. arXiv: 1710.09412[cs,stat]. URL: <http://arxiv.org/abs/1710.09412> (visited on 08/22/2024).
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